

Supplementary Material: Spectral Transformation for Layer-wise Global Rank Discovery in Federated LoRA for Vision Transformers

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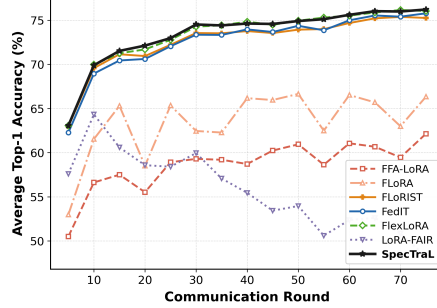
A Additional Experimental Results

A.1 Convergence Across All Model–Dataset Combinations

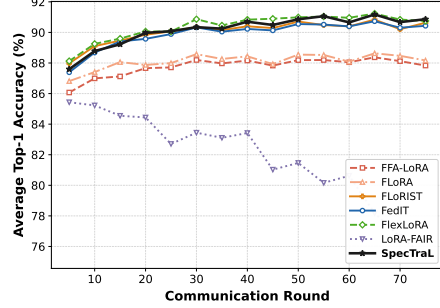
Figure 1 reports training convergence under homogeneous rank settings for all four model–dataset combinations, complementing the DomainNet results shown in the main text (§4.2). Across every combination, SpecTraL tracks the strongest baselines (FlexLoRA, FLoRIST, FedIT) throughout training and converges at least as fast, while the constrained baselines, FFA-LoRA, FLoRA, and LoRA-FAIR, trail consistently. Notably, LoRA-FAIR degrades over training on NICO++, consistent with its inability to fully correct aggregation bias under heterogeneity.

A.2 Additional Initialization Ablation on NICO++

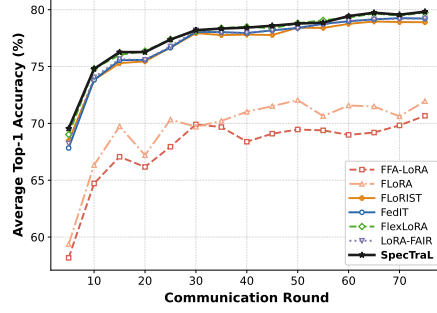
Table 1 reports the initialization ablation on NICO++ (ViT-B/16, $r = 32$, 75 rounds), complementing the DomainNet results in §4.3. The trends observed in the main text hold here as well: Gaussian and Pretrained-SVD remain the two strongest initializers, with Pretrained-SVD reaching the highest average accuracy (91.08%) and Gaussian a close second (90.86%). Zero-padding is again consistently the weakest (89.86%), reinforcing that exploratory directions in the padded dimensions are essential when ScreeNOT yields low-rank global adapters. All strategies except zero-padding exceed FLoRIST (90.63%). Across both datasets, Gaussian ranks among the top two while requiring no access to W_0 or stale local state, supporting its use as the default in SpecTraL.



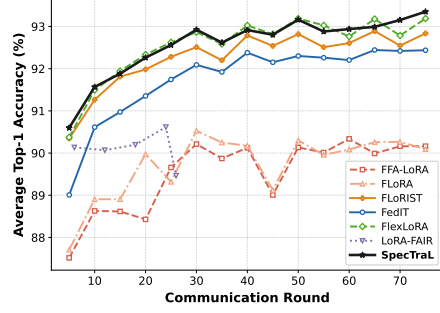
(a) ViT-B/16, DomainNet



(b) ViT-B/16, NICO++



(c) ViT-L/16, DomainNet



(d) ViT-L/16, NICO++

Fig. 1: Training convergence under homogeneous rank settings ($r = 32, 100$ clients, 75 rounds) across all four model–dataset combinations. SpecTraL converges at least as fast as the strongest baselines in every setting, while FFA-LoRA, FLoRA, and LoRA-FAIR trail throughout.

Table 1: Ablation of initialization strategies with ScreenNOT thresholding on NICO++ (ViT-B/16, $r = 32$, 75 rounds). Best and second-best results are **bolded** and underlined.

Initialization	Autumn	Dim	Grass	Outdoor	Rock	Water	Avg.
Zero-padding	90.76	86.86	93.26	89.81	89.15	89.29	89.86
Gaussian	<u>91.58</u>	88.19	93.65	<u>90.86</u>	<u>89.92</u>	<u>90.96</u>	<u>90.86</u>
Trained-A	91.20	<u>88.91</u>	<u>93.48</u>	90.68	89.74	90.60	90.77
Pretrained-SVD	91.75	89.27	93.63	90.62	90.60	90.62	91.08
Orthogonal compl.	90.81	88.79	93.97	90.53	90.19	90.23	90.75